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I build *humanist physical intelligence* for the human body.

Striving to enable people to live healthier, more fulfilling lives through life-saving, enriching, physically-intelligent systems.

• CARDIOVASCULAR & ORTHOPEDIC MECHATRONICS • SOFT ROBOTICS • ACCESSIBILITY TECH • EMBEDDED AI/ML • CLOUD/FULL-STACK/IOS DEV

Y YALE '28 | MECHANICAL ENGINEERING (ABET) & EECS (DOUBLE MAJOR)

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I'm a Yale junior double-majoring in **ABET Mechanical Engineering** and the **EECS (Electrical Engineering + Computer Science)** joint major. I'm an entrepreneur and biomechatronics & cardiovascular engineer, focused on **building humanist healthcare and physical intelligence through robotics, materials, and ML**. Particularly interested in cardiovascular engineering, but also work extensively in orthopedics and prosthetics. I originate from Italy 🇮🇹, Spain 🇪🇸, & Costa Rica 🇨🇷. Over the years, I've founded several software- and hardware-based ventures and conducted extensive research at leading laboratories worldwide. My broader interests span marathon running, linguistics (polyglot), Stoic philosophy, writing, and theoretical physics.

Education

B.S. in ABET Mechanical Engineering & EECS (Electrical Engineering + Computer Science)
Yale University · New Haven, CT, USA
2024 – 2028

Work Experience

Biomedical Engineering Intern: Biomechanics and Biomaterials · **May 2026 – Present**
Harvard Medical School / Massachusetts General Hospital

- Harris Orthopaedics Laboratory: developing next-generation implant materials, devices, and simulations for orthopedic surgery
- Reducing post-operative infection risk through mechanical devices leveraging pharmacokinetic/pharmacodynamic (PK/PD) optimization
- Simulating intra-articular (in joint) conditions for local antibiotic delivery

Cardiovascular Biomechanics and Tissue Engineering, Undergraduate Researcher · **Feb 2025 – Present**
Yale Integrative Cardiac Biomechanics Laboratory

- Engineering Dynamic Electromechanical Bioreactor (DEB) replicating exercise stress on human-engineered heart tissues (hEHTs)
- Studying Arrhythmogenic Cardiomyopathy progression in endurance athletes
- Developing programmable electromechanical loading system with real-time force feedback + electrophysiological readouts

President · **Aug 2024 – Present**
Yale Undergraduate Robotics

- First sophomore ever selected to lead a venture at Yale's largest engineering society
- Directing 30+ engineers across mechanical, robotics, science, and E&C subteams for URC 2026 (Utah)
- Tech: ROS2, sensor fusion, LiDAR, ML/TensorFlow, SLAM, pose estimation, computer vision
- Leading design reviews, test campaigns (mobility, manipulator, bio payloads), system-level risk mitigation

Summer Research Fellow, Reactor Engineering · **Jun – Aug 2025**
UCLouvain – Université catholique de Louvain

- Simulated CO_2 methanation reactors under concentrated streams; evaluated methane yield & thermal runaway thresholds
- Built heterogeneous multibed + multitubular fixed-bed reactor models
- Applied CFD, MATLAB, Python for dynamic simulations; improved parameter estimation via model reduction
- Delivered technical reports on flue gas cleaning + energy efficiency under Prof. Juray De Wilde

Robotics Engineer · **Apr 2025 – Present**
The Faboratory at Yale (Kramer-Bottiglio Lab)

- Bio-inspired burrowing robots for planetary exploration and environmental sensing
- Designing soft robotic actuators, locomotion tests, control systems for granular media navigation

Mechanical Design Engineer, Precision Microtome Device · **Spring 2026**
Johnson & Johnson Vision Care

- Selected to design and build a precision microtome-style device capable of producing ~10 μm reproducible sections of soft, compliant polymers without embedding or cryogenic stiffening
- Developing custom clamping, micron-scale feed mechanisms, and blade kinematics to minimize compression, shear, and deformation during sectioning
- Engineering a non-contaminating mechanical sectioning workflow to enable accurate refractive-index depth profiling across the first 100–1000 μm of material
- Validating section thickness accuracy, flatness, and repeatability using optical microscopy and mechanical calibration methods

Lead, UCT Algorithmic Benchmarking · Aug 2024 – Sept 2025
US Space Command / Yale Space Policy & Research Collaborative

- Spearheaded Yale's collaboration with U.S. Department of Defense (Space Command) on space situational awareness
- Led cross-disciplinary research on satellite UCT algorithms coordinating government, academia, industry
- Results accepted at AMOS 2025 (Maui, Hawaii), premier global space domain awareness conference

Mobile App Engineer · Sept 2025 – Present
Yale Department of Political Science / Government of Greece

- Mobile app for Greek government aiding migrants/asylum seekers: legal rights, hotlines, job matching
- Cross-platform stack (React Native, TypeScript) serving thousands of end-users across Europe
- Bridging Yale research with real-world humanitarian impact

Platform Engineer · Sept 2024 – Aug 2025
CourseTable

- One of ~10 Yale undergrads selected; most-used website at Yale (~95% student body, millions of semester requests)
- Engineered professor analytics insight page, optimized calendar view, high-reliability features

Frontend Systems & Architecture Engineer · May – Aug 2024
VoqoAI Australia

- Led frontend systems design for B2B AI calling platform backed by Australia's largest investors
- Built core features from scratch (agent management, call logs) from Figma to full-stack
- Distributed team collaboration across 16-hour time difference; React, TypeScript, Next.js, MongoDB, Docker

Platform Engineer · Dec 2022 – Jan 2023
Aviato · Part-time · Remote

- One of the first five employees; company now valued at $9M+
- Built crucial early features in a fast-paced startup environment with VC backing and online virality

Ventures & Startups

Founder · Apr 2025 – Present
DegreeIntelligence

- Yale's leading degree-tracking platform, used by 1 in 7 undergraduates (~750+ users)
- Built full-stack entirely solo: Next.js, Firebase, OpenAI APIs
- AI advising + collaborative planning features

Co-Founder, Lead Engineer (Acquired) · Apr – Nov 2023
HSC-GPT → Acquired by Art of Smart Education

- Australia's first localized AI education platform for national curriculum (HSC)
- Adopted by thousands of students, outpacing edtech incumbents
- Led engineering of AI-powered adaptive learning system; acquired 2023

Founder & Lead Engineer · Jul 2022 – Jan 2024
Diush

- Open-source Gen-Z social commerce app; 5,000+ users on waitlist
- Raised VC funding from 1517 Fund (Medici Grant)
- Won 1st place in AASCA international entrepreneurship competition (Central America)
- Drove design, engineering, and growth as sole maintainer

CEO & Co-Founder · Jan – Nov 2021
Zyndicate

- All-in-one productivity ecosystem ("LifeOS") for entrepreneurs and teams
- Scaled to 31,500+ users (10,500 weekly active) by age 16
- Recruited and led 15+ member international team across 8 countries
- VC conversations with Andreessen Horowitz; explored acquisition paths

Co-Founder · Jan – Nov 2021
UpDrop Drone Systems (World Robot Olympiad)

- At age 15, co-engineered autonomous drone delivery system for WRO Open Category
- 1st place Costa Rica, 2nd worldwide out of 14,000+ teams
- Featured in international media (La Nación, Teletica, El Mundo)
- Commended by President of Costa Rica with personal mention and public statement

Creator · Mar 2023 – Present
YouTube Channel: "Filippo Studies"

- 200,000+ total views, 1,500+ subscribers
- Content: learning systems, productivity frameworks, polyglot strategies (7+ languages)
- Featured: "19-year-old polyglot speaks in 7 languages", "Reading ALL THE ESSAYS that got me into Yale"

Papers & Publications

Uncorrelated Track (UCT) Processing: Establishing Efficient Algorithmic Treatment and Benchmarking Best Practices for Space Domain Awareness (SDA) · **Published**
AMOS Conference 2025 · Maui, Hawaii
F. Fonseca, O. Pinhasi, Z. Zitzewitz · Paper · September 2025

Standardizing Experimental Setups for Robotic Locomotion in Granular Media · **Published**
IEEE RoboSoft 2026 · Japan
C.L. Le, **F. Fonseca***, N. Ramos*, E.I. Figueroa, C. Creager, R. Kramer-Bottiglio · Paper · April 2026

Awards

- Y **Y Combinator Startup School 2026** – selected as one of a few builders out of 30,000+ applicants worldwide; attending in San Francisco (July 2026)
- **Google x Yale SOM Hackathon 2026** – 2nd Place Overall with iSpy, a wearable AI vision system for the blind and visually-impaired
- **YHack 2026** – 5th Place Hardware at Yale's premier national hackathon with PosturePoke, a posture wearable that fights slouching with escalating punishments
- **World Robot Olympiad 2021** – 2nd Place Worldwide (14,000+ teams); 1st Place Costa Rica
- **1517 Fund Medici Grant** – VC grant recipient
- **HSC-GPT Acquisition** – Platform acquired by Art of Smart Education (2023)
- **AASCA Entrepreneurship** – 1st Place, international competition (Central America)

Technical Skills

Engineering	Mechatronics · Robotics · Soft Robotics · Reactor Modeling · CFD · PCB Design · Power Systems
Software	TypeScript · Python · React/Next.js · Firebase · MATLAB · C/C++ · Java · Flutter · Rust · Git · Docker · ROS2
AI/ML	TensorFlow · PyTorch · Computational Modeling · Data Analysis · Regression · Predictive Modeling · OpenAI APIs
Hardware	SolidWorks · Fusion 360 · Arduino · Raspberry Pi · Embedded Systems · PCB Design · 3D Printing
Leadership	Product Management · Technical Writing · Team Leadership (15+ reports) · Cross-Disciplinary Collaboration

Languages

🇺🇸 **English** · Native 🇮🇹 **Italian** · Native 🇪🇸 **Spanish** · Native 🇫🇷 **French** · Fluent 🇧🇬 **Bulgarian** · Conversational 🇧🇷 **Portuguese** · Conversational 🇳🇴 **Norwegian** · Learning